GenAI for Engineers Lesson Plan

IES | v2.0 | Three connected Excel labs

Latest user-directed scope supersedes the broader original six-module/16-lab allocation. The course title and IES provider remain. Focus: use Copilot in Excel for data quality, SPC, FMEA and DMAIC.

Outcomes

1 Audit engineering data and verify Copilot formulas. 2 Build and interpret a fixed-baseline SPC chart. 3 Calculate and critically review an FMEA. 4 Produce an exposure-aware DMAIC summary and control plan. 5 Use clear prompts, constraints and evidence-based review.

Time	Block	Delivery focus
09:00–09:30	Copilot setup and prompt framework	30 min
09:30–11:15	Lab01 SPC: quality audit, formulas and baseline	105 min
11:15–11:30	Tea	Break
11:30–12:05	Lab01 SPC: chart and interpretation	35 min
12:05–13:00	Lab02 FMEA: structure and ratings	55 min
13:00–14:00	Lunch	Break
14:00–14:45	Lab02 FMEA: prioritisation and review	45 min
14:45–15:15	Lab03 DMAIC: define and measure	30 min
15:15–15:30	Tea	Break
15:30–16:30	Lab03 DMAIC: analyse, improve and control	60 min
16:30–17:00	Workbook presentations and review	30 min

390 instructional minutes:330 lab minutes +30 opening +30 review. Lunch60 minutes and tea30 minutes. Total09:00–17:00.

Delivery and resources

Three self-contained lab folders, each with Starter.xlsx, Worked-Reference.xlsx, prompts, CSV evidence and PDF handout. Use pairs but each learner saves evidence. Confirm Copilot Excel entitlement and approved storage. A manual formula fallback is allowed but must not be reported as a Copilot execution.

Assessment of learning

Formative review: correct source rows and denominators, correct formulas, chart interpretation, preserved uncertainty and traceable next action. No extra lab list or agent deployment is required.